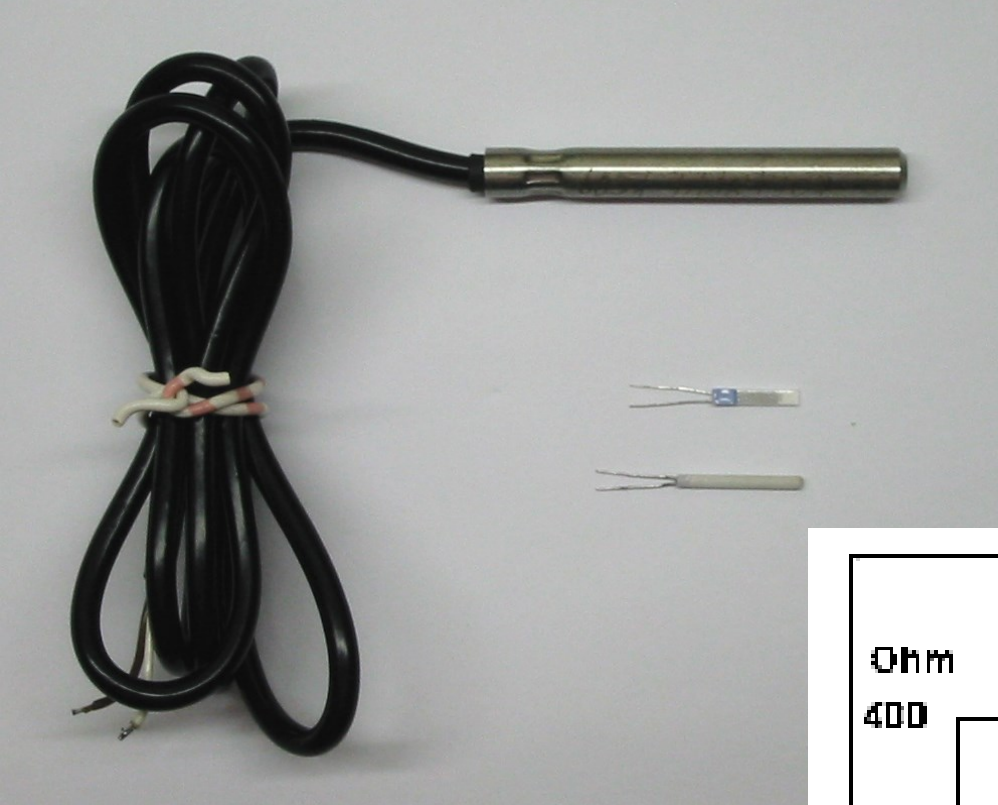


Resistive Temperature Detectors (RTD)

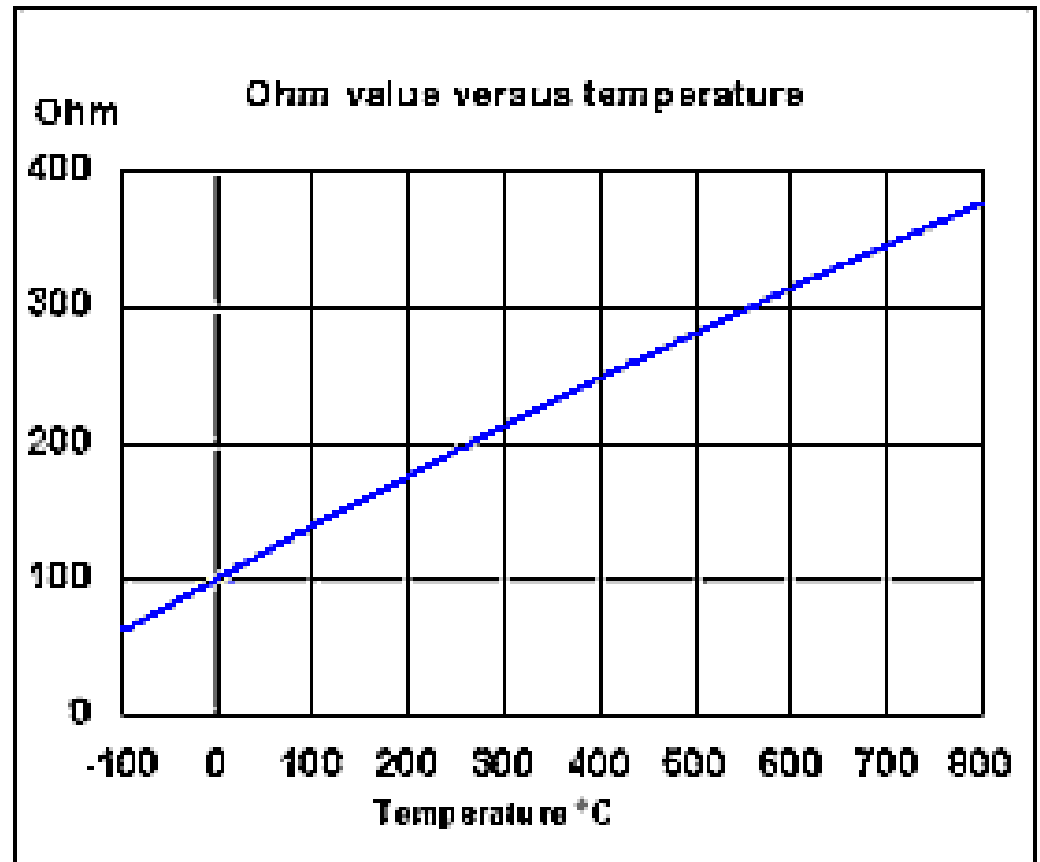
<i>Material</i>	<i>Temperature Range C (F)</i>	<i>Typical Resistance (ohms)</i>
<i>Platinum</i>	<i>-200 - 850 (-328 - 1562)</i>	<i>100</i>
<i>Nickel</i>	<i>-80 - 300 (-112 - 572)</i>	<i>120</i>
<i>Copper</i>	<i>-200 - 260 (-328 - 500)</i>	<i>10</i>

0.4 ohms/°C.

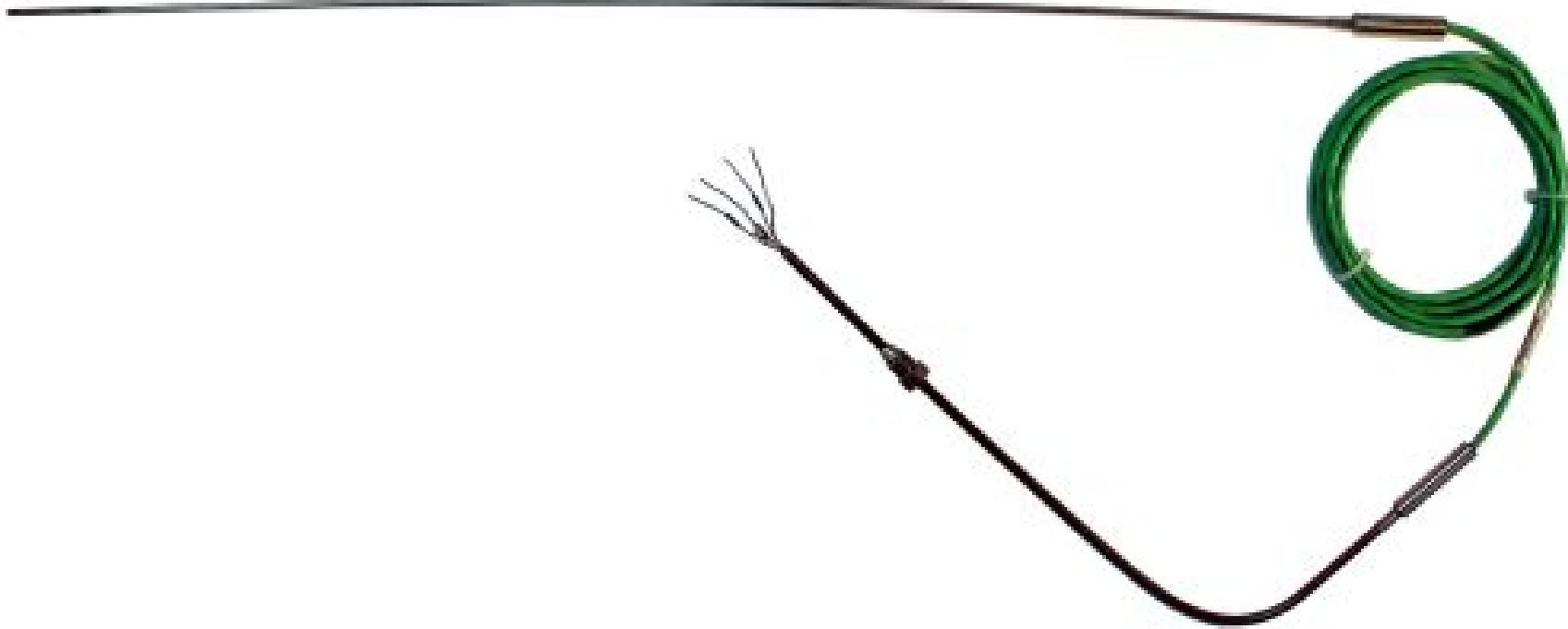
Measurement of voltage drop at a known, constant, small current.



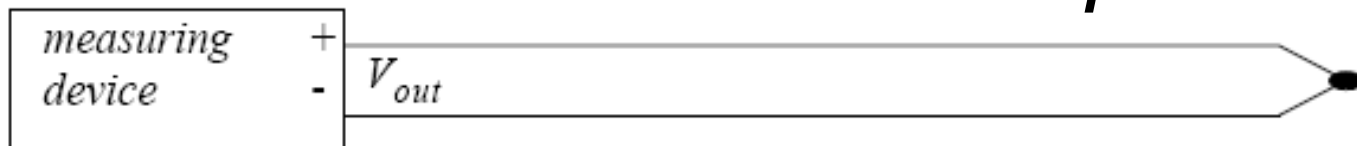
PT-100



Thermocouples



Thermocouples



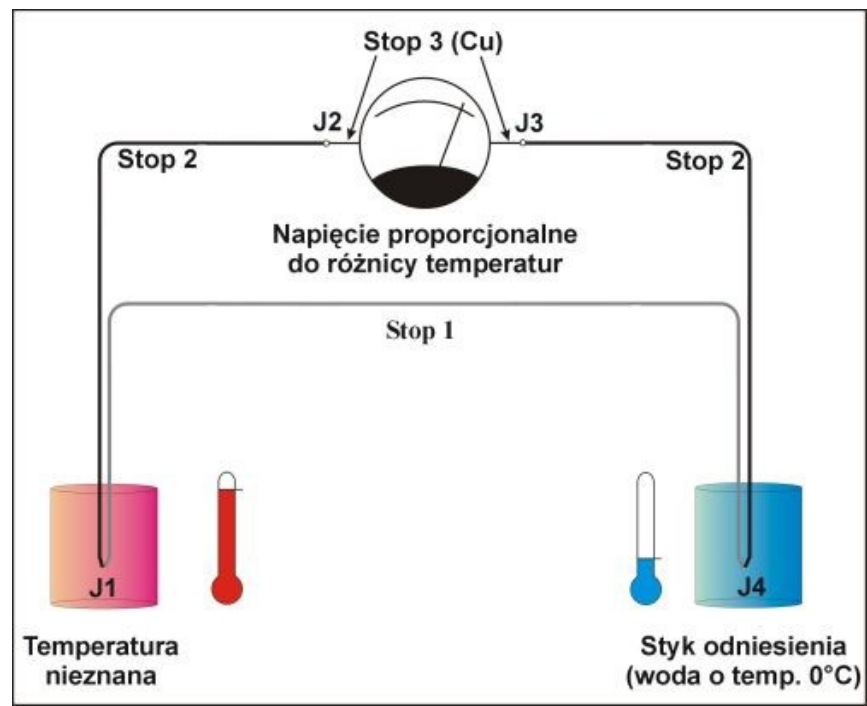
$$V_{out} = \alpha(T - T_{ref})$$

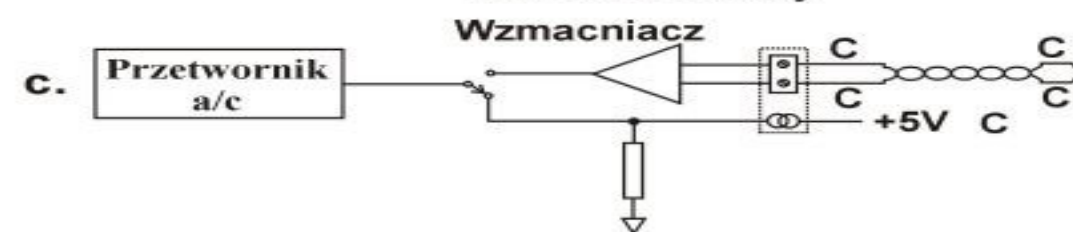
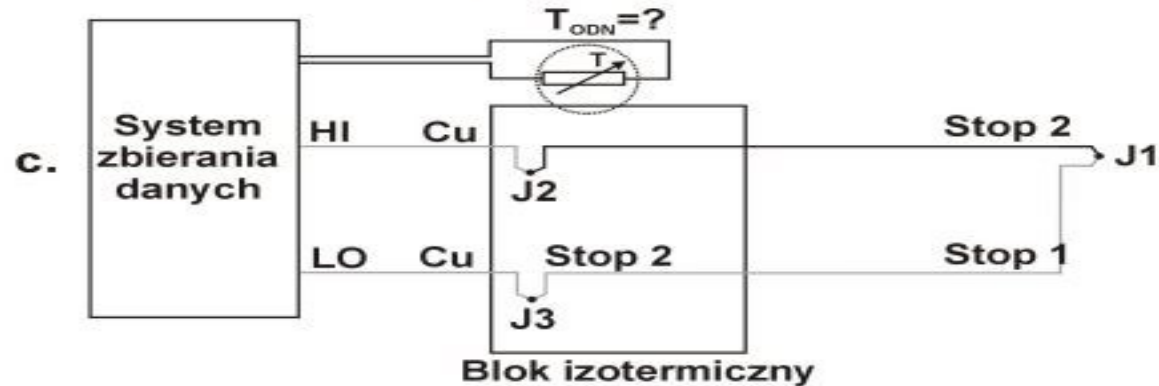
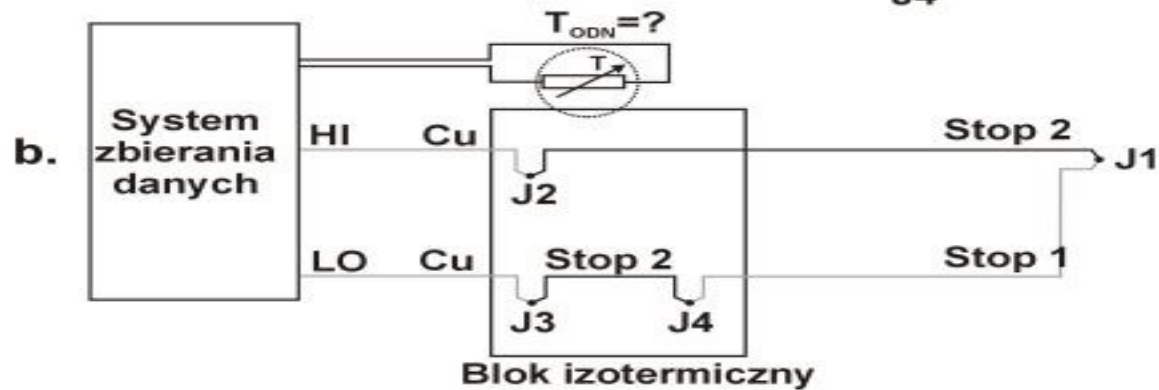
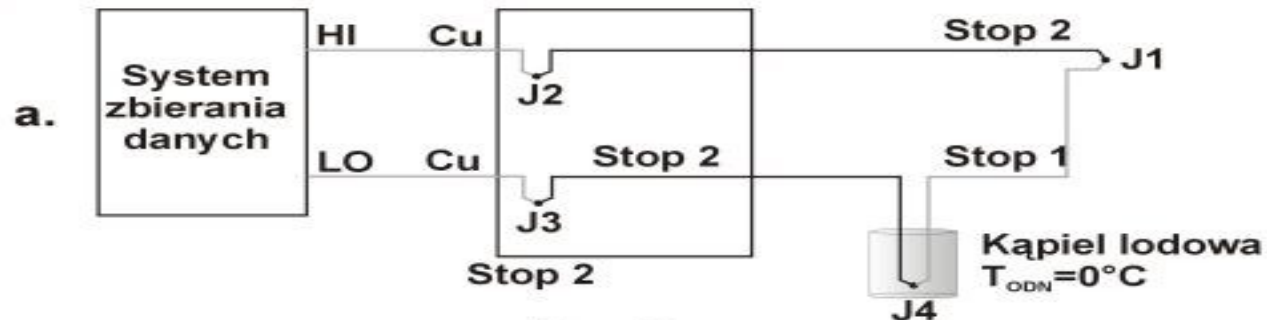
$$\therefore T = \frac{V_{out}}{\alpha} + T_{ref}$$

where,

$$\alpha = \text{constant (V/C)} \quad 50 \frac{\mu V}{^{\circ}C} \text{ (typical)}$$

T, T_{ref} = current and reference temperatures

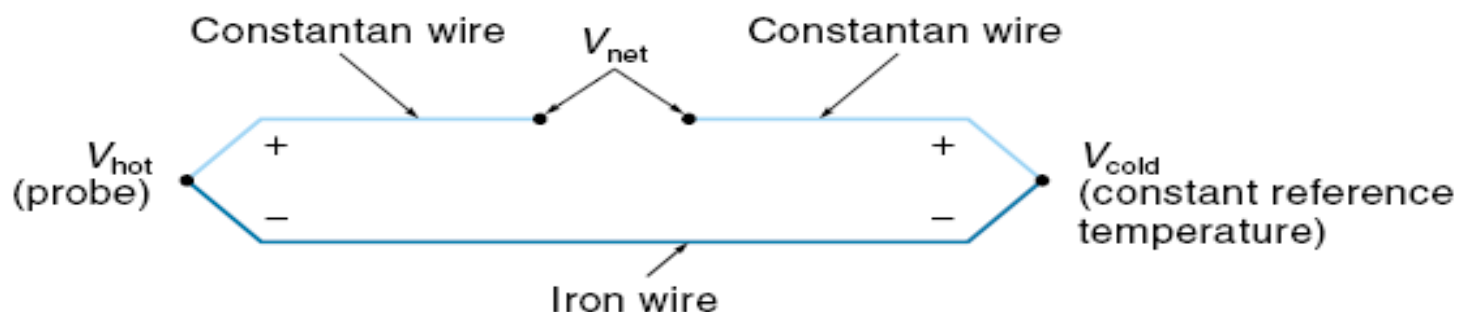


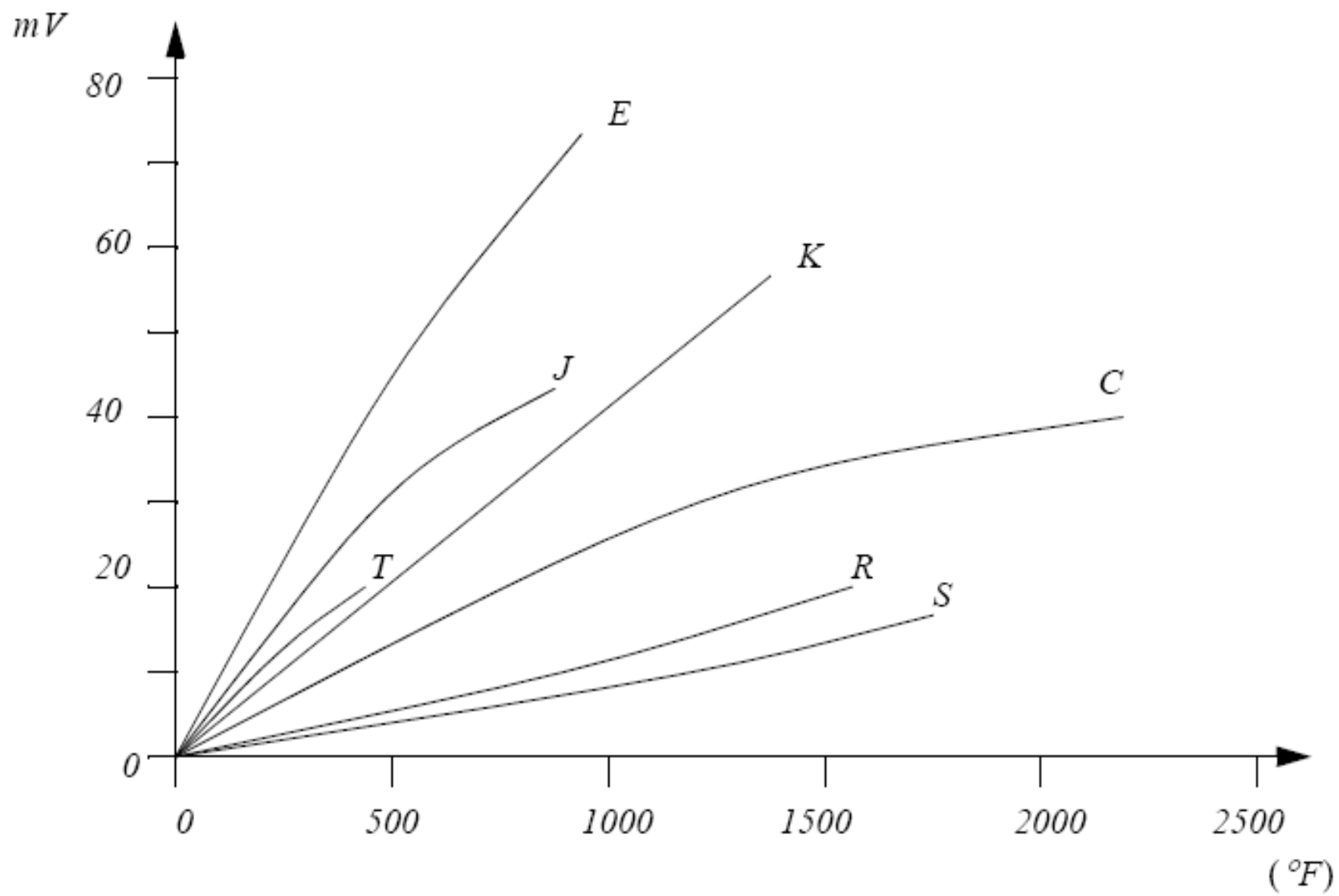


Typowe wejście termopary typu J wykorzystującej półprzewodnikowy czujnik temperatury jako styk odniesienia.

Table 1: Thermocouple Types

ANSI Type	Materials	Temperature Range (°F)	Voltage Range (mV)
T	copper/constantan	-200 to 400	-5.60 to 17.82
J	iron/constantan	0 to 870	0 to 42.28
E	chromel/constantan	-200 to 900	-8.82 to 68.78
K	chromel/aluminum	-200 to 1250	-5.97 to 50.63
R	platinum-13%rhodium/platinum	0 to 1450	0 to 16.74
S	platinum-10%rhodium/platinum	0 to 1450	0 to 14.97
C	tungsten-5%rhodium/tungsten-26%rhodium	0 to 2760	0 to 37.07





Thermistors

Negative temperature coefficient (NTC)

$$R_t = R_o e^{\beta \left(\frac{1}{T} - \frac{1}{T_o} \right)}$$

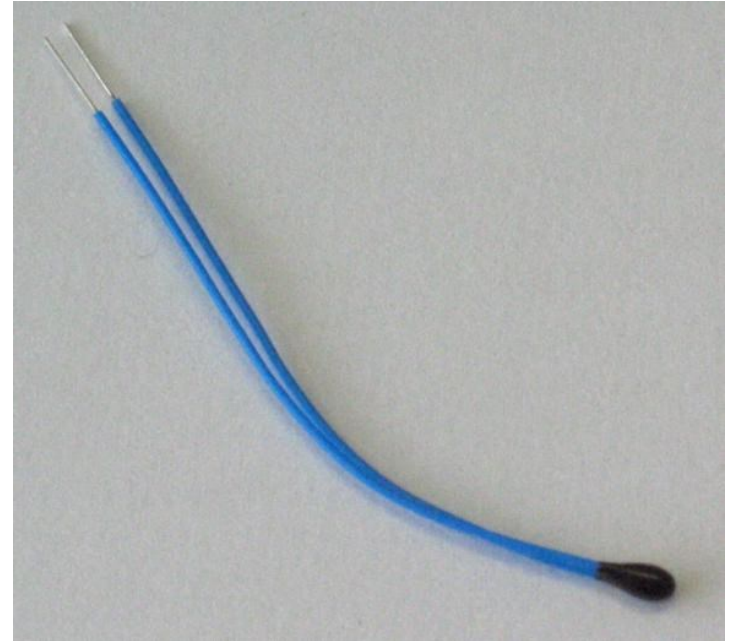
$$\therefore T = \frac{\beta T_o}{T_o \ln \left(\frac{R_t}{R_o} \right) + \beta}$$

where,

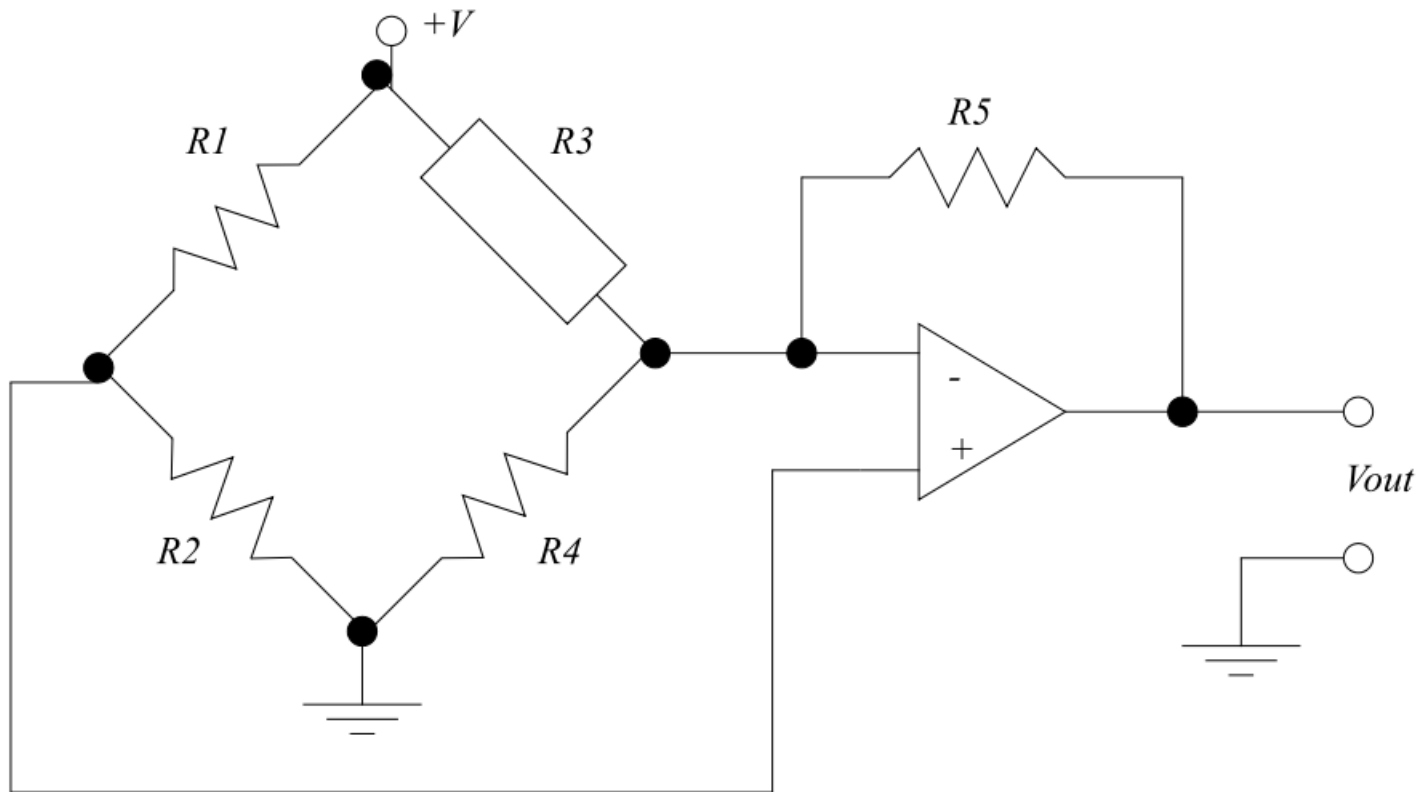
R_o, R_t = resistances at reference and measured temps.

T_o, T = reference and actual temperatures

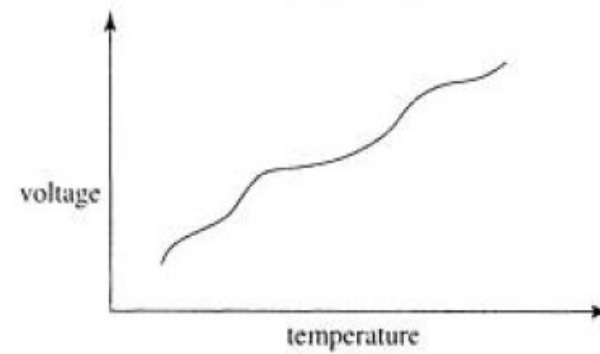
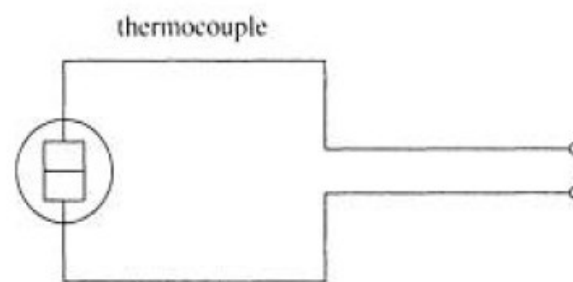
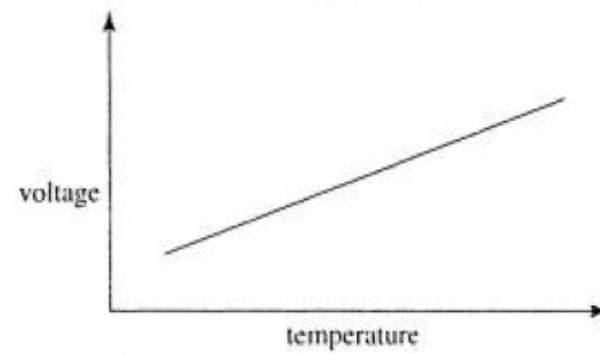
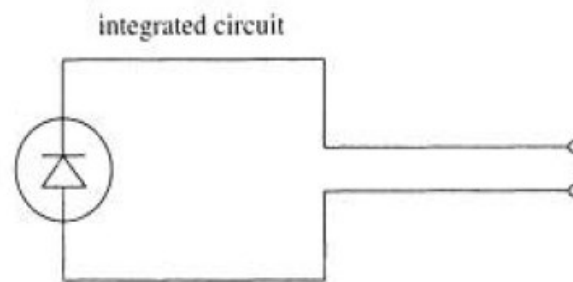
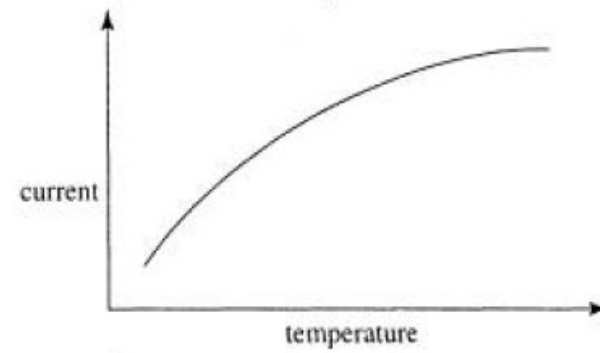
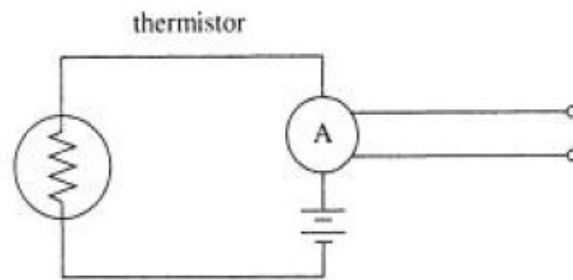
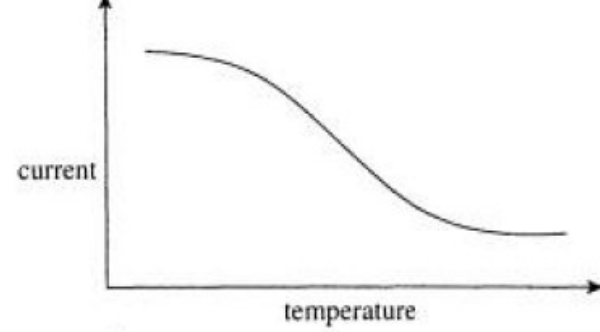
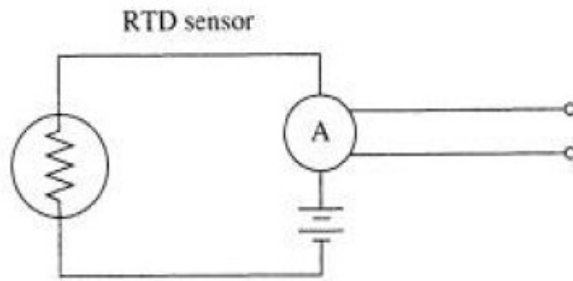
β = constant for device



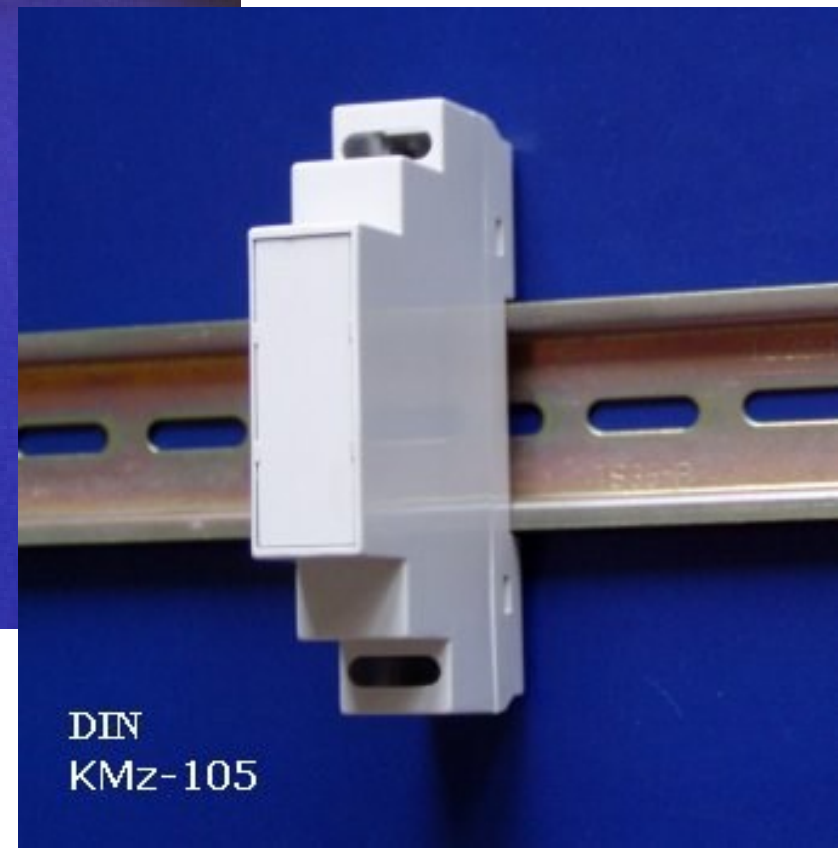
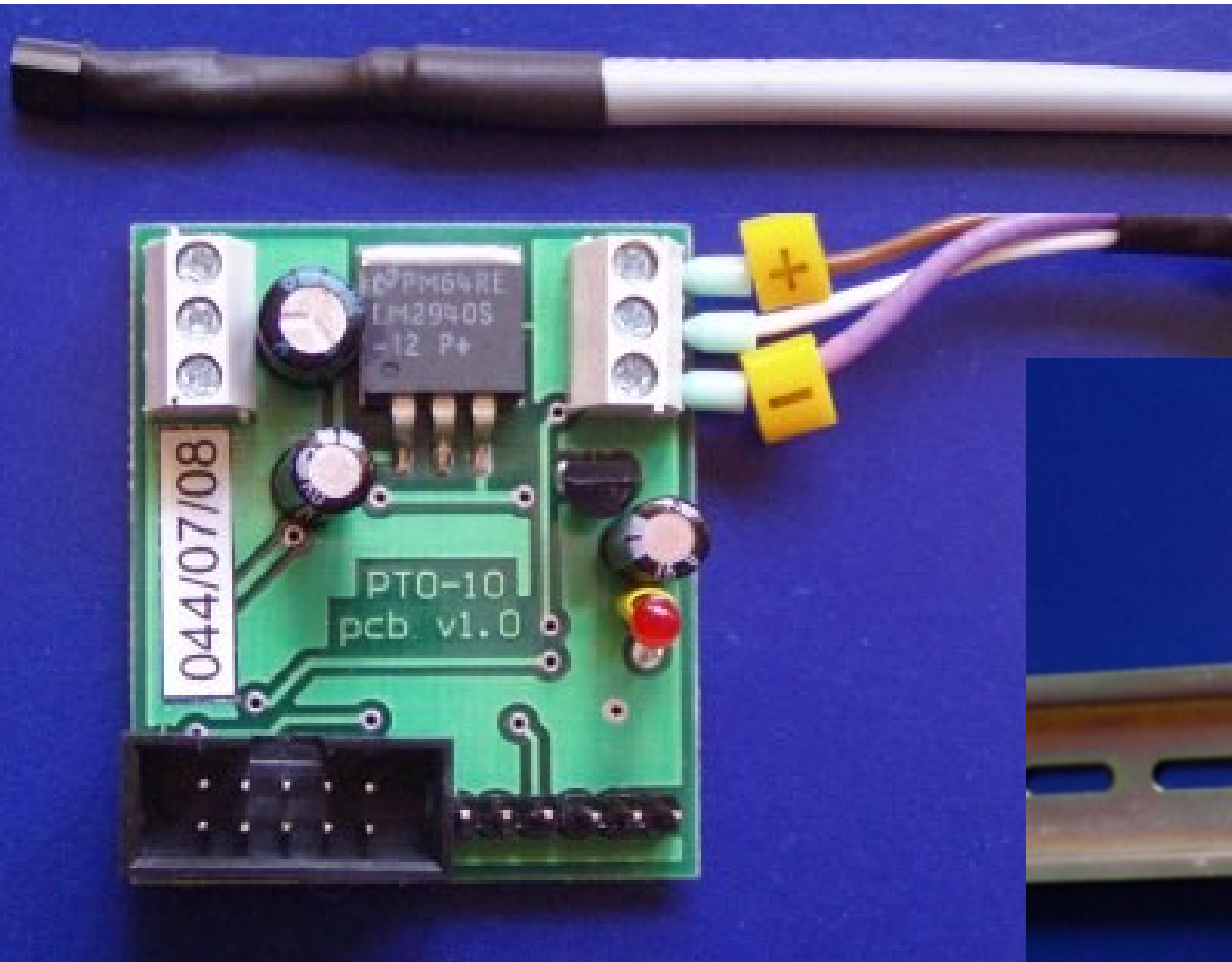
Thermistors



Typical : 1, 10, 100 k Ω , -50 st. C do 120 st C



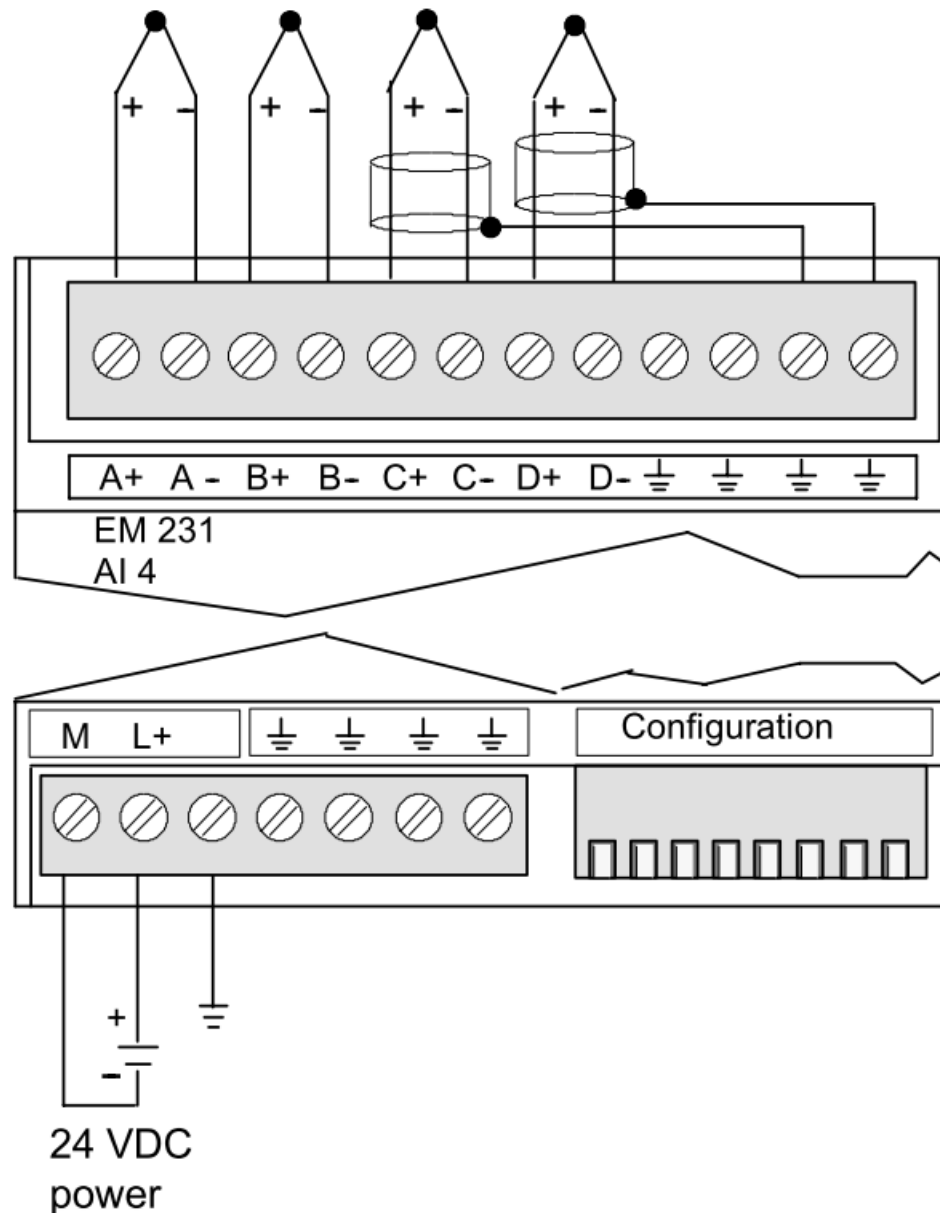
PT0-10 sensor based on DS18B20



PT0-10 Sensor

- Operating temperature limits: -50 st C, +120 st C
- Measurement resolution: 0.1 degree
- Accuracy of measurement: 0.5 degrees
- Output signal : 0-10 V
- Output load capacity: up to 50 mA
- Short-circuit protection: yes
- Supply voltage: 12-24 V
- Dimensions: transmitter 36x42mm, sensor 5mm
- 2 m three-core cable, which can be extended up to 20 m

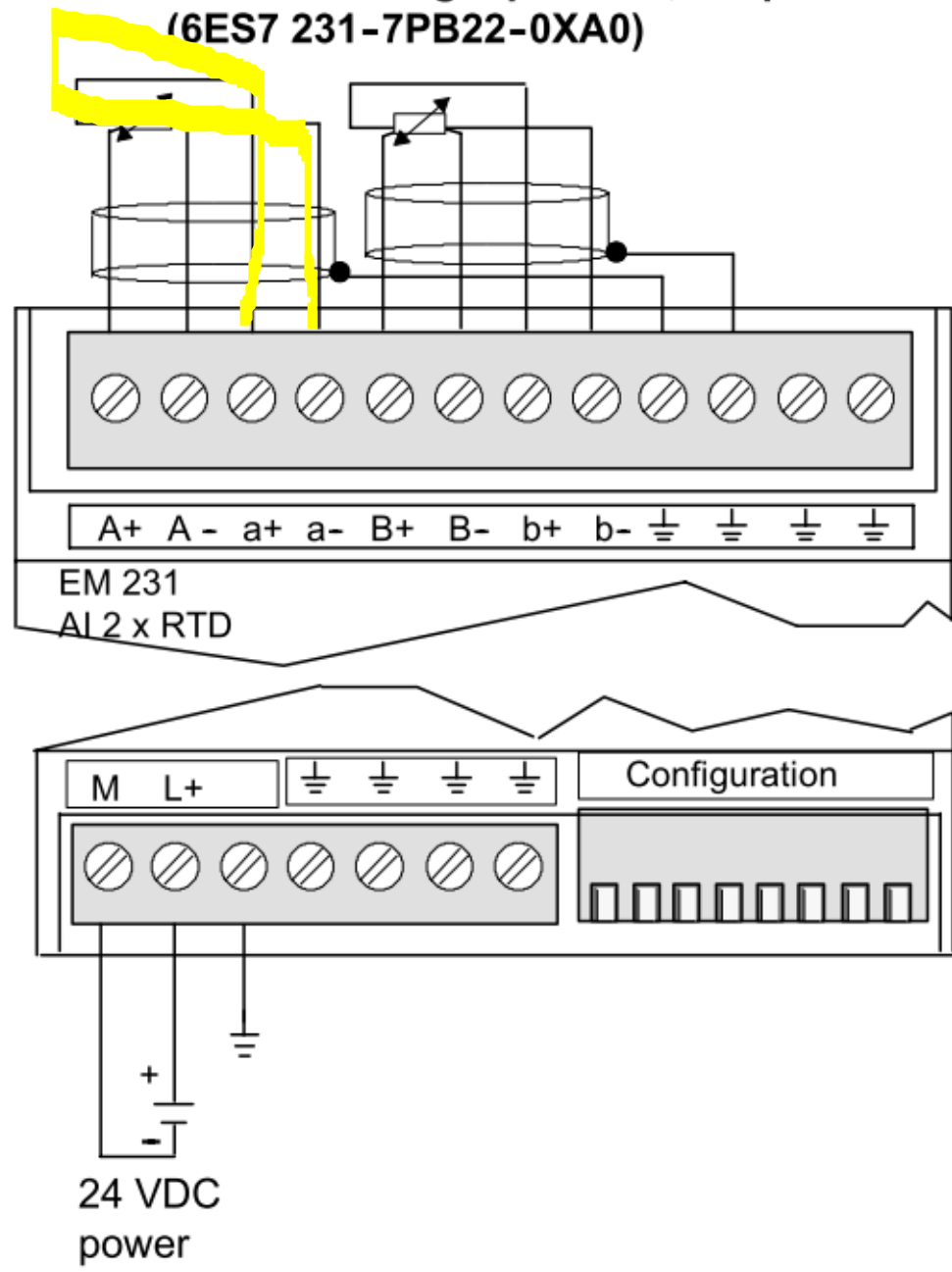
EM 231 Analog Input Thermocouple, 4 Inputs (6ES7 231-7PD22-0XA0)



	Thermocouple
Isolation	
Field to logic	500 VAC
Field to 24 VDC	500 VAC
24 VDC to logic	500 VAC
Common mode input range (input channel to input channel)	120 VAC
Common mode rejection	> 120 dB at 120 VAC
Input type	Floating TC
Input ranges ¹	TC types (select one per module) S, T, R, E, N, K, J Voltage range : +/- 80 mV
Input resolution	
Temperature	0.1° C / 0.1° F
Voltage	15 bits plus sign
Resistance	-
Measuring Principle	Sigma-delta
Module update time: All channels	405 ms
Wire length	100 meters to sensor max.
Wire loop resistance	100Ω max.
Suppression of interference	85 dB at 50 Hz/60 Hz/ 400 Hz
Data word format	Voltage: -27648 to + 27648
Maximum sensor dissipation	-
Input impedance	≥1 MΩ
Maximum input voltage	30 VDC
Input filter attenuation	-3 db at 21 kHz
Basic error	0.1% FS (voltage)
Repeatability	0.05% FS
Cold junction error	±1.5 ° C
24 VDC supply voltage range	20.4 to 28.8 VDC

Data Word (1 digit = 0.1 °C)		Type J	Type K	Type T	Type E	Type R, S
Dec	Hex					
32767	7FFF	>1200.0 °C	>1372.0 °C	>400.0 °C	>1000.0 °C	>1768.0 °C
↑	↑					
32511	7EFF					
:	:					
27649	6C01					
27648	6C00					↑
:	:					
17680	4510		↑			1768.0 °C
:	:					
13720	3598		1372.0 °C			
:	:		overrange			
13000	32C8	↑	1300.0 °C			
:	:					
12000	2EE0	1200.0 °C			↑	
:	:					
10000	2710			↑	1000.0 °C	
:	:					
4000	0FA0			400.0 °C		400.0 °C
:	:					
1	0001	0.1 °C	0.1 °C	0.1 °C	0.1 °C	0.1 °C
0	0000	0.0 °C	0.0 °C	0.0 °C	0.0 °C	0.0 °C
-1	FFFF	-0.1 °C	-0.1 °C	-0.1 °C	-0.1 °C	-0.1 °C
:	:					underrange
-500	FE0C					-50.0 °C
-1500	FA24	-150.0 °C				↓
:	:					
-2000	F830	underrange	-200.0 °C			
:	:					
-2100	F7CC	-210.0 °C				
:	:					
-2400	F6A0		underrange		-240.0 °C	
:	:				underrange	
-2550	F60A			-255.0 °C		
:	:			underrange		
-2700	F574	↓	-270.0 °C	-270.0 °C	-270.0 °C	
:	:					
-27648	9400		↓	↓	↓	
-27649	93FF					
:	:					
-32512	8100					
↓	↓					
-32768	8000	<-210.0 °C	<-270.0 °C	<-270.0 °C	<-270.0 °C	<-50.0 °C
Accuracy over full span		±0.1%	±0.3%	±0.6%	±0.3%	±0.6%
Accuracy (normal range without cold junction)		±1.5 °C	±1.7 °C	±1.4 °C	±1.3 °C	±3.7 °C
Cold junction error		±1.5 °C	±1.5 °C	±1.5 °C	±1.5 °C	±1.5 °C

EM 231 Analog Input RTD, 2 Inputs (6ES7 231-7PB22-0XA0)

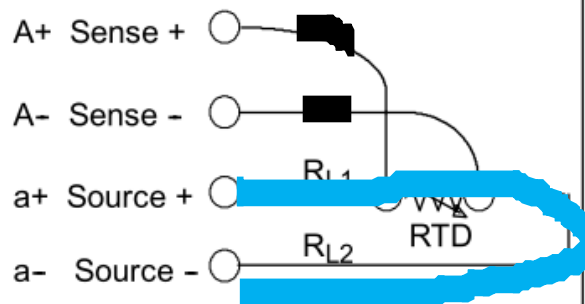


	RTD
Isolation	
Field to logic	500 VAC
Field to 24 VDC	500 VAC
24 VDC to logic	500 VAC
Common mode input range (input channel to input channel)	0
Common mode rejection	> 120 dB at 120 VAC
Input type	Module ground referenced RTD
Input ranges ¹	RTD types (select one per module): platinum (Pt), copper (Cu), nickel (Ni), or Resistance See Table A-30 for available RTD types.
Input resolution	
Temperature	0.1° C / 0.1° F
Voltage	-
Resistance	15 bits plus sign
Measuring Principle	Sigma-delta
Module update time: All channels	405 ms (700 ms for Pt10000)
Wire length	100 meters to sensor max.
Wire loop resistance	20Ω, 2.7Ω for Cu max.
Suppression of interference	85 dB at 50 Hz/60 Hz/400 Hz
Data word format	Resistance: 0 to +27648
Maximum sensor dissipation	1m W
Input impedance	≥ 10 MΩ
Maximum input voltage	30 VDC (sense), 5 VDC (source)
Input filter attenuation	-3 db at 3.6 kHz
Basic error	0.1% FS (resistance)
Repeatability	0.05% FS
Cold junction error	-
24 VDC supply voltage range	20.4 to 28.8 VDC

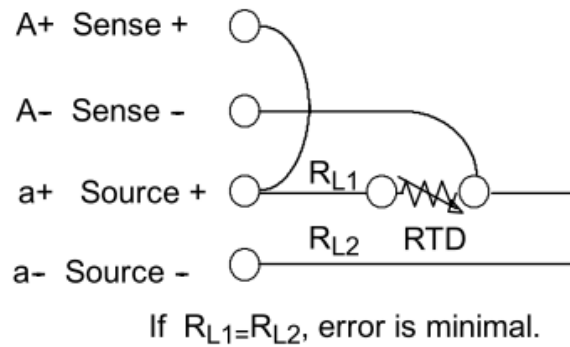
RTD Type and Alpha ¹	SW1	SW2	SW3	SW4	SW5	RTD Type and Alpha ¹	SW1	SW2	SW3	SW4	SW5
100Ω Pt 0.003850 (Default)	0	0	0	0	0	100Ω Pt 0.003902	1	0	0	0	0
200Ω Pt 0.003850	0	0	0	0	1	200Ω Pt 0.003902	1	0	0	0	1
500Ω Pt 0.003850	0	0	0	1	0	500Ω Pt 0.003902	1	0	0	1	0
1000Ω Pt 0.003850	0	0	0	1	1	1000Ω Pt 0.003902	1	0	0	1	1
100Ω Pt 0.003920	0	0	1	0	0	SPARE	1	0	1	0	0
200Ω Pt 0.003920	0	0	1	0	1	100Ω Ni 0.00672	1	0	1	0	1
500Ω Pt 0.003920	0	0	1	1	0	120Ω Ni 0.00672	1	0	1	1	0
1000Ω Pt 0.003920	0	0	1	1	1	1000Ω Ni 0.00672	1	0	1	1	1
100Ω Pt 0.00385055	0	1	0	0	0	100Ω Ni 0.006178	1	1	0	0	0
200Ω Pt 0.00385055	0	1	0	0	1	120Ω Ni 0.006178	1	1	0	0	1
500Ω Pt 0.00385055	0	1	0	1	0	1000Ω Ni 0.006178	1	1	0	1	0
1000Ω Pt 0.00385055	0	1	0	1	1	10000Ω Pt 0.003850	1	1	0	1	1
100Ω Pt 0.003916	0	1	1	0	0	10Ω Cu 0.004270	1	1	1	0	0
200Ω Pt 0.003916	0	1	1	0	1	150Ω FS Resistance	1	1	1	0	1
500Ω Pt 0.003916	0	1	1	1	0	300Ω FS Resistance	1	1	1	1	0
1000Ω Pt 0.003916	0	1	1	1	1	600Ω FS Resistance	1	1	1	1	1

¹ All RTDs represent 0° C. at the listed resistance except for Cu 10 ohm. Cu 10 ohm is 25° C. at 10 ohm and 0° C. at 9.035 ohm.

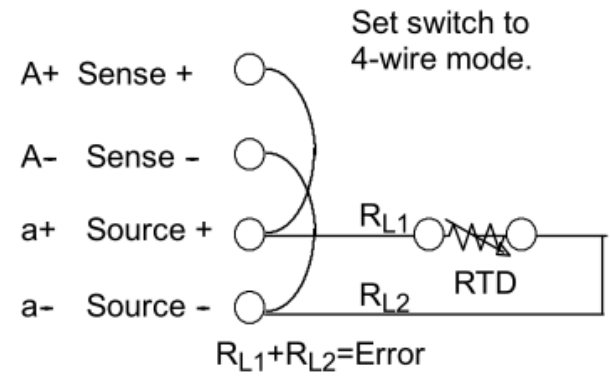
RTD 4 Wire (most accurate)



RTD 3 Wire



RTD 2 Wire



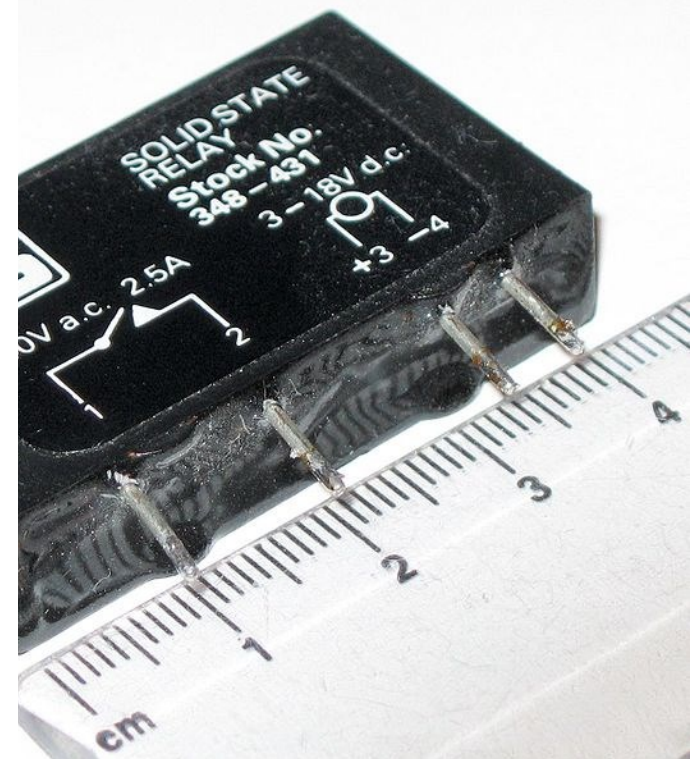
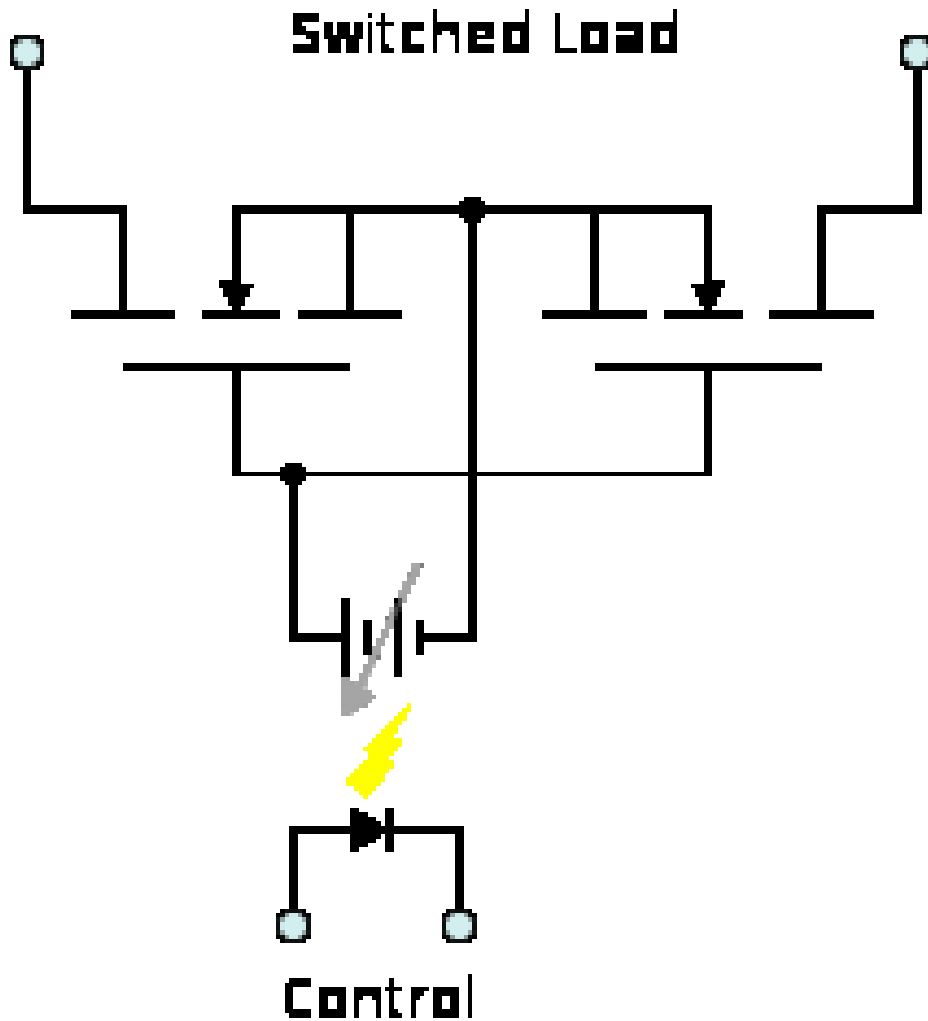
System Word (1 digit = 0.1 °C)		Pt10000	Pt100, Pt200, Pt500, Pt1000	Ni100, Ni120, Ni1000 ¹	Cu10		
Decimal	Hex						
32767	7FF.						
32766	7FFE						
32511	7EFF						
29649	6C01						
27648	6C00						
25000	61A8						
18000	4650						
15000	3A98						
13000	32C8	↑	↑				
10000	2710	1000.0°C	1000.0°C				
8500	2134		850.0°C				
6000	1770	600.0°C			↑		
3120	0C30			↑	312.0°C		
2950	0B86			295.0°C	260.0°C		
2600	0A28						
2500	09C4			250.0°C			
1	0001	0.1°C	0.1°C	0.1°C	0.1°C		
0	0000	0.0°C	0.0°C	0.0°C	0.0°C		
-1	FFFF	-0.1°C	-0.1°	-0.1°C	-0.1°C		
-600	FDA8			-60.0°C			
-1050	FBE6			-105.0°C	-200.0°C		
				↓			
-2000	F830	-200.0°C	-200.0°				
-2400	F6A0				-240.0°C		
-2430	F682	-243.0°C	-243.0°C				
		↓	↓				
-5000	EC78						
-6000	E890						
-10500	D6FC						
-12000	D120						
-20000	4E20						
-32767	8001						
-32768	8000						
Accuracy over full span		±0.4%	±0.1%	±0.2%	±0.5%		
Accuracy (normal range)		±4° C	±1° C	±0.6° C	±2.8° C		



Main

Range of product	TeSys D
Product or component type	Contactor
Device short name	LC1D
Contactor application	Motor control Resistive load
Utilisation category	AC-1 AC-3
Control circuit type	DC
Coil type	Low consumption
Poles description	3P
Pole contact composition	3 NO
Control circuit voltage	24 V DC

SSR *Solid-state relay*





Electronic contactors



- unlimited service life.
- No sparking during switching.
- No contact burnout or interference.
- No heat generation.
- no voltage drop when conducting, which occurs with SSR relays.
- No need to use heat sinks.
- Built-in varistor protection.
- Life span is limited to the mechanical life of the relay: approximately 100 million times.